

Abstract Algebra

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1. Sets

1.0.1. Set theory

Definition 1.1 (Tuple)

A tuple (or sometimes **list**) is a finite, ordered collection of elements, possibly with repetitions.

An n -tuple is of length n . A 2-tuple is called an **ordered pair**.

1.0.2. Morphisms

Definition 1.2 (Map)

Subsets of the Cartesian product $A \times B$ are called **relations**. A **map** is a **well-defined** relation written

$$f \subseteq A \times B \quad \text{or} \quad f : A \rightarrow B \quad \text{or} \quad A \xrightarrow{f} B$$

and is

- surjective** or **onto** if $f(A) = B$, which is equivalent to being **right invertible** with $f \circ f^{-1} = \text{id}_B$.
- injective** or **one-to-one** if $a_1 \neq a_2 \implies f(a_1) \neq f(a_2)$, which is equivalent to being **left invertible** with $f^{-1} \circ f = \text{id}_A$.
- bijective** or a **one-to-one correspondence** if both above apply, which is equivalent to being **invertible**.

Definition 1.3 (Homomorphism)

A **homomorphism** is a map between algebraic structures that preserves the structure.

A group homomorphism $\varphi : G \rightarrow H$ satisfies

$$\varphi(g * \gamma) = \varphi(g) \cdot \varphi(\gamma)$$

for all $g, \gamma \in G$.

Theorem 1.1

If $\varphi : G \rightarrow H$ is a group homomorphism,

- (i) $\varphi(\text{id}_G) = \text{id}_H$
- (ii) $\varphi(g^{-1}) = \varphi(g)^{-1}$ for all $g \in G$

Definition 1.4 (Isomorphism)

An **isomorphism** is a bijective homomorphism.

If there exists an isomorphism between two algebraic structures, they are said to be **isomorphic**, denoted with $\cong$.

Definition 1.5 (Endomorphism)

An **endomorphism** is a homomorphism from a mathematical object to itself.

Definition 1.6 (Automorphism)

An **automorphism** is simultaneously an isomorphism and an endomorphism: a bijective homomorphism from an object to itself.

Example. The **identity map** id_X is a trivial automorphism defined

$$\text{id}_X(x) = x \quad \forall x \in X.$$

Example. An automorphism of a set is called a **permutation**.

1.0.3. Equivalence

Definition 1.7 (Equivalence Relation)

An **equivalence relation** on a set A is a relation $R \subseteq A \times A$ with the following properties:

- (i) **reflexive** $(a, a) \in R \quad \forall a \in A$
- (ii) **symmetric** $(a, b) \in R \implies (b, a) \in R$
- (iii) **transitive** $(a, b), (b, c) \in R \implies (a, c) \in R$

If $(x, y) \in R$ we write $x \sim y$.

Definition 1.8 (Partition)

A **partition** $\mathcal{P}$ of a set A is a collection of non-empty, disjoint subsets of A whose union is A .

Definition 1.9 (Equivalence Class)

Let R be an equivalence relation on a set A . The **equivalence class** of $a \in A$ is

$$[a] = \{b \in A \mid b \sim a\}.$$

The set of all equivalence classes is written

$$A // R$$

and is called the **quotient set**.

The equivalence classes form a partition of A .

Example. The modulo m congruence class $[a] = \{a + km \mid k \in \mathbb{Z}\}$.

Definition 1.10 (Quotient Map)

Let R be an equivalence relation on A . The map $\pi : A \rightarrow A // R$ defined by

$$\pi(a) = [a]$$

is called the **quotient map**.

It is always surjective.

1.0.4. Linear Diophantine Equations

Let a and b be integers with $\text{gcd}(a, b) = d$.

Lemma (Bézout's Identity)

There exist integers x and y such that $ax + by = d$.

Lemma (Euclid's)

If p is prime and $p \mid ab$, then $p \mid a$ or $p \mid b$.

Theorem 1.2 (Solvability Condition)

The linear Diophantine equation $ax + by = c$ has (infinitely many) solutions if and only if $d \mid c$.

Theorem 1.3 (Solutions)

The solutions, if they exist, are

$$x = x_0 + t \frac{b}{d}, \quad y = y_0 - t \frac{a}{d}, \quad \text{for } t \in \mathbb{Z}.$$

Theorem 1.4 (Euclidean Algorithm)

Let a and b be non-negative integers with $a \geq b$. Then there exist unique integers q and r such that

$$a = qb + r, \quad 0 \leq r < b.$$

Example. We apply the algorithm to compute $\text{gcd}(252, 105)$:

$$\begin{aligned} 252 &= 2 \cdot 105 + 42 \\ 105 &= 2 \cdot 42 + 21 \\ 42 &= 2 \cdot 21 + 0 \end{aligned}$$

Hence, $\text{gcd}(252, 105) = 21$.

Example. We then use our results to solve $252x + 105y = 21$.

Since $\text{gcd}(252, 105) = 21$, we now work backwards:

$$\begin{aligned} 21 &= 105 - 2 \cdot 42 \\ &= 105 - 2 \cdot (252 - 2 \cdot 105) \\ &= 105 - 2 \cdot 252 + 4 \cdot 105 \\ &= 5 \cdot 105 - 2 \cdot 252 \\ &= -2 \cdot 252 + 5 \cdot 105 \end{aligned}$$

So a particular solution is $x_0 = -2, y_0 = 5$. The general solutions are

$$x = -2 + 5t, \quad y = 5 - 12t, \quad \text{for } t \in \mathbb{Z}.$$

2. Groups

2.0.1. Definition

Definition 2.1 (Group)

A **group** $(G, *)$ is a nonempty set G with a binary operation

$$* : G \times G \rightarrow G$$

such that for all $a, b, c \in G$, it satisfies

- (i) (Closure) $a * b \in G$
- (ii) (Associativity) $(a * b) * c = a * (b * c)$
- (iii) (Identity) $\exists e \in G : e * a = a * e = a$
- (iv) (Inverse) $\exists a^{-1} \in G : a * a^{-1} = a^{-1} * a = e$

Additionally, a group is called **abelian** or **commutative** if it is

- (v) (Commutative) $a * b = b * a$

Examples. The **general linear group** $GL_n(\mathbb{R})$ consists of all matrices in $\mathbb{R}^{n \times n}$ with $\det \neq 0$. The **special linear group** $SL_n(\mathbb{R})$ additionally requires that $\det = 1$.

💡 Tip

The proof of (iii) and (iv) can be shortened if (v) is proven.

💡 Note

Defying conventional wisdom, **abelian** is usually not capitalized.

2.0.2. Subgroups

Let $(G, *)$ be a group.

Definition 2.2 (Subgroup)

A subset $H \subseteq G$ is a **subgroup** if it is itself a group under the operation inherited from G .

💡 Tip

It follows that H is a subgroup if

- (i) $H \neq \emptyset$
- (ii) $a * b^{-1} \in H \quad \forall a, b \in H$

Definition 2.3 (Cyclic Subgroup)

For any $a \in G$, the set

$$\langle a \rangle = \{a^k \mid k \in \mathbb{Z}\}$$

forms a **cyclic subgroup** of G .

Theorem 2.1

The cyclic subgroup $\langle a \rangle \leq G$ is the smallest subgroup containing a .

Definition 2.4 (Generator)

An element $a \in G$ is a **generator** of G if $\langle a \rangle = G$.

Definition 2.5 (Cyclic Group)

We say G is a **cyclic group** if it contains a **generator** of itself.

Theorem 2.2

All subgroups of a cyclic group are cyclic.

Theorem 2.3

Every cyclic group is abelian.

Proof. Let g and γ be elements of a cyclic group $G = \langle a \rangle$. Then there are integers m and n such that $g = a^m$ and $y = a^n$, so

$$g\gamma = a^m a^n = a^{m+n} = a^{n+m} = a^n a^m = \gamma g.$$

2.1. COSETS

Let H be a subgroup of a group G .

Definition 2.6 (Coset)

For any $g \in G$, the **left coset** of H in G is

$$gH = \{g * h \mid h \in H\}.$$

Similarly, the **right coset** is

$$Hg = \{h * g \mid h \in H\}.$$

Theorem 2.4

The left and right cosets respectively partition G .

Theorem 2.5

There are equally many left and right cosets of H in G .

Definition 2.7 (Order)

The **order of a group** $|G|$ is the number of elements in G .

The **order of an element** $|g|$ is the smallest positive integer n such that $g^n = e$. If there is no such n , the order is infinite.

Definition 2.8 (Index)

The **index** of H in G is the number of left cosets, denoted $[G : H]$.

Theorem 2.6 (Lagrange's)

Let G be finite. Then the order of H divides the order of G .

Corollary

If G is of prime order, it

- (i) lacks proper nontrivial subgroups
- (ii) is generated by any element $g \neq e$
- (iii) is cyclic

Theorem 2.7 (Lagrange's, Generalized)

Regardless of whether G is finite or infinite,

$$|G| = [G : H] |H|.$$

Proof. G has $|G|$ elements and is partitioned into $[G : H]$ distinct left cosets, each of which has $|H|$ elements.

Definition 2.9 (Normal Subgroup)

The following are equivalent for a subgroup H of G :

- (i) H is a **normal subgroup** of G , denoted $H \triangleleft G$.
- (ii) $ghg^{-1} \in H$ for all $g \in G$ and $h \in H$.
- (iii) $gHg^{-1} \subseteq H$ for all $g \in G$.
- (iv) $gHg^{-1} = H$ for all $g \in G$.
- (v) $gH = Hg$ (equal left coset and right coset) for all $g \in G$.
- (vi) The sets of left and right cosets coincide.

Examples. Trivially, G and $\{e\}$ are always normal subgroups.

Definition 2.10 (Simple Group)

A group is **simple** if it has no normal nontrivial proper subgroup.

Theorem 2.8

All subgroups of abelian groups are normal subgroups.

2.2. RINGS

2.2.1. Definition

Definition 2.11 (Ring)

A **ring** is a set R with two binary operations $+$ and $\cdot$ such that

- (i) $(R, +)$ is an abelian group with identity 0.
- (ii) $(R, \cdot)$ is a monoid with identity 1.
- (iii) Multiplication distributes over addition:
for all $a, b, c \in R$, $a \cdot (b + c) = a \cdot b + a \cdot c$.

Definition 2.12 (Group of Units)

Let $(R, +, \cdot)$ be a ring. The **group of units** $U(R)$ is the set of all elements in R that have multiplicative inverses, that is

$$U(R) = \{a \in R \mid \exists a^{-1} \in R : a \cdot a^{-1} = a^{-1} \cdot a = 1\}.$$

2.2.2. $\mathbb{Z}_n$

Let a and n be positive integers and let p be prime.

Definition 2.13 ($\mathbb{Z}_n$)

The **integers modulo n** is a set of congruence classes

$$\mathbb{Z}_n = \{[0], [1], \dots, [n-1]\}$$

that form an abelian group under $+$ and a monoid under $\times$, together forming a ring with a group of units called the

multiplicative group of integers modulo n , denoted $U(n)$ and consisting of the integers coprime to n , with inverse elements defined to satisfy $aa^{-1} \equiv 1 \pmod{n}$.

Definition 2.14 (Euler's Totient Function)

$$\phi(n) = |U(n)| = |\{k \in \mathbb{Z} \mid 1 \leq k \leq n \text{ and } \text{gcd}(n, k) = 1\}|$$

Example. $\phi(12) = |\{1, 5, 7, 11\}| = 4$

Theorem 2.9 (Euler's)

If a and n are coprime, then

$$a^{\phi(n)} \equiv 1 \pmod{n}.$$

Proof. The order of a is $|U(n)| = \phi(n)$.

Lemma

$$\phi(p) = p - 1$$

Theorem 2.10 (Fermat's Little)

If a and p are coprime, then

$$a^{p-1} \equiv 1 \pmod{p}.$$

In general, even if they are not coprime,

$$a^p \equiv a \pmod{p}.$$

2.3. PERMUTATIONS

2.3.1. Transpositions

Definition 2.15 (Transposition)

A transposition of the elements $i, j \in \{1, \dots, n\}$ is the permutation τ_{ij} with the cyclic notation (ij) .

Theorem 2.11

A permutation $\sigma \in S_n$ can be written as a composition of transpositions.

Theorem 2.12

The identity $\text{id} \in S_n$ may only be written as an even number of transpositions.

Definition 2.16 (Sign of a Permutation)

Let k be the number of transpositions in any decomposition of a permutation $\sigma \in S_n$. The **sign** of σ is $(-1)^k$. Hence,

- if k is **even** then $\text{sgn}(\sigma) = +1$ and σ is said to be even.
- if k is **odd** then $\text{sgn}(\sigma) = -1$ and σ is said to be odd.

Definition 2.17 (Fixed Point)

A fixed point is invariant under a transformation such as a permutation.

2.3.2. Symmetric Groups

Definition 2.18 (Symmetric Group)

The **symmetric group** of the finite set $X = \{1, 2, \dots, n\}$ is

$$S_n = \{\sigma : X \rightarrow X \mid \sigma \text{ is bijective}\}$$

and consists of $n!$ permutations, which form **the symmetric group on n letters** under the operation of composition

Theorem 2.13

The automorphism group of a set is its symmetric group.

Definition 2.19 (Permutation Group)

A subgroup of a symmetric group S_n is a **permutation group**.

Definition 2.20 (Alternating Group)

The **alternating group**

$$A_n = \{\sigma \in S_n \mid \text{sgn}(\sigma) = 1\}$$

is the subgroup of S_n consisting of all even permutations.

It has $n!/2$ elements for $n \geq 2$.

Theorem 2.14

The sign map $\text{sgn} : S_n \rightarrow \{1, -1\}$ is a group homomorphism.

The kernel of this homomorphism is the alternating group

$$\ker(\text{sgn}) = A_n.$$

2.4. SYMMETRY

Definition 2.21 (Isometry)

Let (X, d_X) and (Y, d_Y) be metric spaces. A map $f : X \rightarrow Y$ is an **isometry**